Detection Of Copy-Create Image Forgery Using Luminance Level Techniques

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***Abstract*- The luminance of an image is the measurement of the perceived brightness levels. Intuitively, if two images are taken from different cameras with different lighting, some sort of discrepancy may occur in those areas, which were copied and pasted. In particular, analyzing areas in a forged image which are approximately the same distance away from the lens but have different luminance levels. This analysis is heavily dependent on the skill level of the person creating the forgery and the resources available to perform the manipulation. Newer versions of image processing software make it easy for even a novice user to create forgeries based on automated "auto-brightness" adjustments. We succeeded to identify such image forgeries**

Keywords- Luminance level, forged image, auto brightness level, Image forgery

I. INTRODUCTION

Digital images are everywhere: on the covers of magazines, in newspapers, in courtrooms, and all over the Internet. We are exposed to them throughout the day and most of the time, we trust what we see. But given the ease with which images can be manipulated, we need to be aware that seeing does not always believe.In recent years, tampered images have affected science, law, politics, the media, and business. Some cases have made national and international headlines, tarnishing the public's perception of images. While forgeries are not a new problem, the tools for making forgeries, such as digital cameras, computers, and software, have increased in sophistication, bringing the ability to make forgeries out of the hands of specialists to anyone. The tools for detecting forgeries, on the other hand, are only beginning to be developed. There is a clear need for these tools if the public is to regain trust in published images.

Digital images can be tampered or manipulated in many different ways. The most commonly used methods are Enhancing, composting, copy-move, copy-paste, etc. A tampered image usually presents no visual artifacts and is not perceptible, so we authenticate it using mathematical and computational techniques. Copy-move tampering [4] in which a part of image is copied and pasted somewhere else in the image to cover or add an important image feature is a simple and powerful form of image manipulation. A tampered image usually undergoes a sequence of post processing manipulation for retouching to conceal traces of tamper, such as lossy JPEG compression, Gaussian blur filter, Gaussian noise contamination, etc.

In this paper, we proposed a novel scheme for identifying the location of copy-create tampering and authenticating an Image by applying the luminance level of techniques such as luminance level, HSV and alternative filtering masks. After reviewing related works in Section II, we present basic ideas of luminance level techniques, and details of the proposed method of identifying tampered regions in an image in Section III. The experimental results and system boundary discussion were present in Section IV. Final conclusions are drawn in Section V.

II. PRIOR WORKS

In the literature of digital image forensics, several techniques were proposed to explore different tampering operations, such as re-sampling [2], lighting inconsistencies [3], irregular noise patterns [5], alterations in the correlation introduced by color filter array interpolation [6], estimating the Quantization tables of JPEG image compression [7],[8].Higher order statistics have been proposed to identify contrast Changes, gamma correction [9], and other nonlinear point operations on images [10] and identify photo realistic images [11]. Physics-motivated features have been introduced to distinguish photographic images and computer graphics [12]. A few copy-move image tamper detection schemes have been proposed in the literature [4], [13],[14],[15]. Fridrich etal [4] first investigated the problem of detecting the copy-move tamper and proposed a reliable method based on matching the quantized lexicographically sorted discrete cosine transform (DCT) coefficients of overlapping image blocks to counter this manipulation, the robustness of this method may be improved further. Popescco et al [13] used an alternative representation of blocks based on principal component analysis to identify the similar blocks in the image. The robustness of the method in detecting tampered regions is demonstrated under ranging JPEG compression qualities and additive noise levels. Babak et al [14] proposed a method based on blur moment invariants, its main disadvantage is complexity and computational time. W. Q. Luo et al [15] describe an efficient and robust algorithm for detecting and localizing copy-move malicious tampering.

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In this paper, we first extract the color features along with different threshold values of images, and identify the location of tampered regions which undergone global value threshold, converting images in to HSV and alternative filter mask.

II. LUMINANCE LEVEL TECHNIQUES

A. Detection of tampering based on analysis of Luminance levels

The luminance of an image is the measurement of the perceived brightness levels [15]. Intuitively, if two images are taken from different cameras with different lighting, some sort of discrepancy may occur in those areas, which were copied and pasted. In particular, analyzing areas in a forged image which are approximately the same distance away from the lens but have different luminance levels. This analysis is heavily dependent on the skill level of the person creating the forgery and the resources available to perform the manipulation. Newer versions of image processing software make it easy for even a novice user to create forgeries based on automated "auto-brightness" adjustments. These properties of an image are influenced by the amount and intensity of light when the photograph was taken, the particular camera's representation of color, and any post-processing performed by image manipulation software. In a forged image, these properties may all slightly differ in and around tampered areas. This is because of the multiple images used to create the forgery, each most likely originating from different cameras and environments. While it is necessary for human interpretation when analyzing the results of an image tamper detection algorithm, there are techniques and methods to assist. A color or grayscale image can be converted to a binary image based on some configurable threshold. A binary image is one in which a pixel is either "on" (filled with black) or "off" (empty or filled with white) [15]. The determination of whether a pixel gets an "on" or "off" value is based on a threshold. In the context of this section, the threshold is the luminance level of an image. The luminance level threshold of an image is represented as a decimal number between 0.00 and 1.00. If the threshold value is 0.75, for example, a pixel is assigned as on (or black) if its luminance level is equal to or less than 0.75. An image is returns all black if the threshold level is set to 1.00 or returns all white if the threshold level is set to 0.00. Determining an appropriate threshold for testing is heavily dependent on the type of image being analyzed. A value of approximately 0.50 is a good starting point with subsequent tests performed in both directions. The ultimate goal is to look for results depicting an area of suspected tampering, which are witnessed by unnatural or abnormal luminance levels in an alleged area.

B. Detection of tampering based on Hue-Saturation-Value (HSV) levels

The Hue of a color is best described as the "tint" [15]. Saturation or "shade" is defined as the level of how pure or intense a color is [15]. Value is the level of brightness (luminance) of a color or how light or dark it is [15]. Intuitively, if an area or areas throughout an image are copied and pasted from different sources, the color and brightness, as captured from each respective camera, may be slightly different. Thorough analysis of HSV levels helps to determine this. Further analysis of color images may be performed with respect to Hue-Saturation-Value (HSV) levels. The standard RGB (Red-Green-Blue) color-space of a digital image represents each color a pixel takes on by an amount of the Red, Green, and Blue components. This is the standard way that color digital images are represented on the computer screen. In the HSV color-space, the amount of Hue, Saturation, and Value represents each color. The color-space and luminance levels of an image provide a unique method of analysis. Discrepancies and other anomalies in these attributes indicate image tampering has taken place.

(a) RGB (b) HSV

Figure.1 – Images in the Red-Green-Blue and Hue-Saturation-Value color-spaces

C. Methods Based on Alternative Filtering Masks

Edge detection algorithms, a classical image processing technique, have been analyzed against a number of forged test images [12]. Lukas analyzed these first since edge detection algorithms are a fundamental application to image processing. The edges of an image are extremely significant in many applications since they provide information about the location of objects and their texture, size, and shape. This concept is of interest in forgery detection because image tampering introduces hidden anomalies often associated with a double edge around the tampered objects. This phenomenon occurs because the blurring of space around the tampered objects, in conjunction with the actual edge, forms a double or "ghost" edge.

An edge is defined as areas in the image where the intensity of pixels moves from a low value to a high value or vice versa [13]. This leads into an analysis of first-order operators and their power at detecting discontinuities. First-order operators detect points in the image that are discontinuous by calculating a function of the image, which uses first-order derivatives. There are various convolution masks used in image processing and some have already been used to analyze forged digital images. Previous images were analyzed using the Roberts, Sobel, and Prewitt masks

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[12]. The Sobel mask is more receptive to edges that are diagonal in nature rather than horizontal or vertical. The Roberts mask is more susceptible to noise than the other masks while Prewitt is better at horizontal and vertical edges[13].The following formula computes the convolution of an image [12]:

$$x,y = \sum_{i=d} \sum_{j=d} {}^{g}i,j \, {}^{f}xi,yj \qquad (1)$$

Where $D = \frac{-(S-1)}{2}$ and $n = \frac{S}{2}$, g is a convolution mask of size s x s, and f is the image Function. The following are the masks described above and used for the variable g.

Robert's Mask **Sobel Mask** **Prewitt Mask**

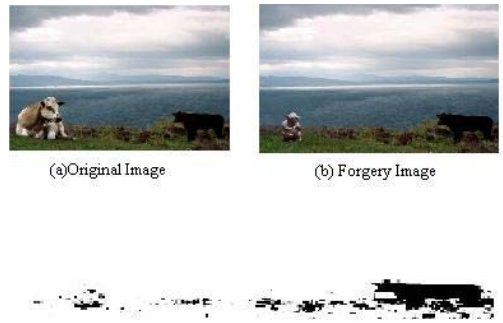
$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

This method of filtering an image is formally classified as pixel-group or spatial domain filtering. While one reason to perform this type of processing is to detect edges, these filtering methods give an alternative way to view an image and therefore may uncover small anomalies introduced from image tampering. The power to create customized masks may prove to be of some interest in detecting image forgeries.

IV. EXPERIMANTAL RESULTS AND SYSTEM BOURNDRY

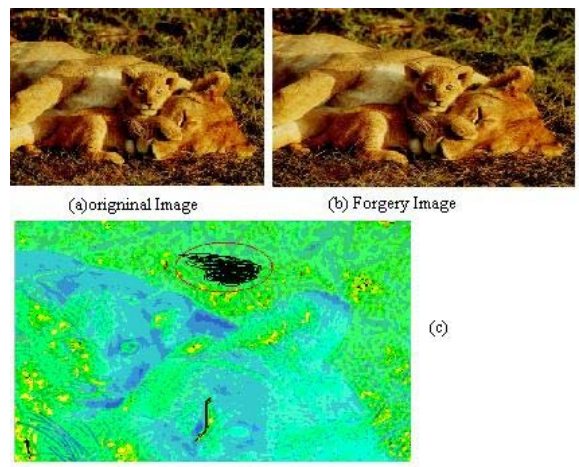
The proposed method has been implemented using Matlab 7.0. The experimental results presented in this section are divided into two parts. In the First part the experiment was conducted with 250 authenticated images along with their forgery has been done with Photoshop and photo edit image manipulation software's. The sample results are shown in (Fig.2 and Fig.3 (b)) presents the reliableness of the luminance technique. In the second Fig4 presents the HSV technique and third part Fig5 presents the Custom convolution mask. In the second part System boundary of proposed methods was discussed.

FIG(2) First Two pictures are original images and third image is copy-created image referingn first two.pic(d) is the result extracted to identified with luminance 0.60



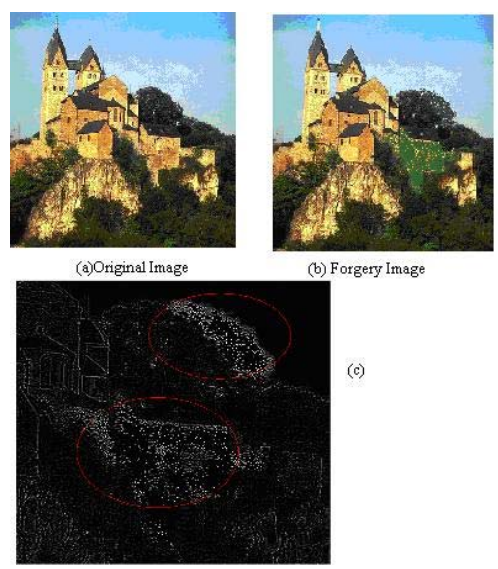
(a)Original Image (b) Forgery Image

Fig 3. (a)Original Image (b) Forgery Image (c) shows the result of forgery identification with 0.15 threshold.



(a)origninal Image (b) Forgery Image (c)

Fig4.(a) Original Image (b) Copy created forgery image (c) Result of HSV technique obtained from fig(b)



(a)Original Image (b) Forgery Image (c)

Fig (5) (a) Original Image (b) Forgery Image (c) Result of custom high pass Filter using fig(b)

All of the methods presented are tailored to the JPEG image format. This format was chosen because of its popularity on

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the web and with digital cameras [1]. Other image formats such as PNG (Portable Network Graphics) and BMP (Windows Bitmap) work with the proposed Luminance and HSV methods as well as with the alternative filtering masks. Obviously, the JPEG Block technique only accurately works with images saved in the JPEG format. The best way to detect image tampering, in general, is to perform a wide range of tests on the image in question. The methodology presented in this chapter attempts to accomplish this by including a broad range of detection techniques. Table 1 includes a listing of the proposed detection methods and those image formats that work with each.

Table 1. Proposed Detection Methods and Tested Image Format

Forgery Detection Method	RGB Color	Gray Scale	JPEG	PNG	BMP
Luminance Level Threshold	X	X	X	X	X
HSV Color-Space	X		X	X	X
Custom High-Pass Filter	X	X	X	X	X

V. CONCLUSION

This paper focuses on methods to detect digital forgeries created from multiple images called as *copy-create* image forgeries. Images that result from portions copied and moved within the same image to "cover-up" something are classified as *copy-move* forgeries. Therefore, the experimental design and analysis herein focuses on *copy-create* image forgeries.

A crafty individual, who wants to perfect an image forgery, with time not a factor, can usually give any detection method trouble. If image tampering occurs in an uncompressed image and then that image is converted to the JPEG image format, the JPEG Block Technique will fail to capture evidence of tampering. This conversion process destroys all proof of tampering since the original tampering does not affect any JPEG blocks. Additionally, any image tampering performed on an image prior to an image size reduction will eliminate detectable anomalies for the custom filtering mask.

By and large, boundaries will always exist in detecting image tampering. Some assumptions need to be made about countermeasures a forger used when crafting an image forgery. The methods presented in this paper are well suited to detect varying skill levels of a person performing image manipulations, but given enough time and resources there are ways to give any detection algorithm difficulty in accurately deciding if an image is authentic.

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